Knife-Throw Arcade Game — Lovable Master Prompt

An Android-only, offline, English knife-throwing arcade game (Capacitor) with real AdMob ads, two currencies (apples and gems), 50 knives and four game modes.

How to use this doc. Paste everything under Master Prompt into Lovable. If Lovable stalls, paste Part 0 + Part 1 first, then Parts 2, 3 and 4, one at a time. Lovable builds the web app only; the native Android steps are in the checklist at the end, and the numbers I assumed for you are under Defaults I chose.

Master Prompt

Part 0 — Product and stack

Build a production-quality, offline, portrait-only Android arcade game in the knife-throwing genre: tap to throw a knife at a rotating target; knives already stuck in it must never be touched; slice apples for currency; every 5th stage is a boss fight that rewards an exclusive knife. The game has four modes (Endless, Fixed Run, Daily Challenge, Time Attack), power-ups, achievements, daily missions, a 7-day Daily Gift and a deep 50-knife store with two currencies (apples and gems). Everything is saved on the device: no login, no backend, no online leaderboard. It ships only on Google Play through Capacitor, so design phone-first (360–430 px wide), respect safe areas and display cutouts, and target 60 fps on mid-range Android phones.

Name and language. Invent an original, catchy, easy-to-pronounce game name that does not resemble any existing game, and use it everywhere (splash, Home logo, Capacitor appName, package id `com.<studio>.<gamename>` in lowercase). Tell me the name and package id you chose. The UI is English only, with every string kept in one `strings.ts` file.

Audience: all ages. Keep the content family-friendly: cartoon style, no blood, no gore, and no people or real animals as targets (targets are logs, food, objects and fantasy creations).

Stack: Vite + React + TypeScript (strict) + Tailwind. Gameplay runs on one HTML Canvas 2D (requestAnimationFrame, fixed-timestep update, delta-time, devicePixelRatio scaling); React renders only menus, HUD and modals. State with Zustand. Save data with Capacitor Preferences and a localStorage fallback, as one versioned save object (apples, gems, owned and equipped knife, best scores per mode, achievements, missions, gift streak, settings, age-screen result). Draw every knife, target, boss, apple and background

procedurally (Path2D / vector code) so no image files are needed. Bundle fonts locally with @fontsource. Audio comes from WebAudio (see Part 4) and must be swappable for real audio files later.

Folders: src/game (engine, entities, stages, bosses, render), src/services (ads, storage, audio, haptics), src/store, src/data (knives.ts, bosses.ts, worlds.ts, achievements.ts, missions.ts), src/ui (screens, components), src/config (ads.ts, adPolicy.ts, economy.ts, difficulty.ts).

Part 1 — Core gameplay

Screens: Splash → age screen (first launch only) → Home → Mode select → Game → Game Over, plus Pause, Boss Intro, Boss Defeated, Level Complete transition, Store, Achievements, Daily Missions, Daily Gift, Daily Challenge and Settings. Add a first-run tutorial (a hand pointer over the tap area for the first three throws).

Core mechanics

- Portrait logical canvas 360×640, scaled to fit. The target (a log) sits upper-center (radius ≈ 80, bosses ≈ 95). The player's knife waits at the bottom; a tap/pointer-down anywhere except on-screen buttons throws it straight up (≈ 1800 px/s), and the next knife appears after ≈ 120 ms. Use touch-action: none and block double-tap zoom.
- Store every stuck knife and hazard as an angle in the target's local frame so they rotate with it. When a thrown knife arrives, compare its angle to all stuck knives and hazards. If the angular gap is smaller than the knife's width at the target radius (≈ 9° for a normal knife, a tunable constant), the throw fails: the knife bounces off and falls, heavy haptic, screen shake, then the fail flow below.
- Apples sit on the target as small angular targets. A knife that hits one slices it (two halves fly off with juice particles), gives the apple's reward, then sticks normally.
- A stage is cleared when all its knives are thrown: the target shatters into pieces with physics, a short Level Complete beat, then the next stage. HUD: knife icons bottom-left (remaining vs used), five stage dots on top with the fifth as a boss skull, apples and gems counters in the top corner, plus in-run points and the power meter (see Power-ups).
- **Fail flow:** a failed run restarts from stage 1. Before restarting, Game Over offers Continue: watch a rewarded ad and resume the same stage exactly where the player stopped (the stuck knives stay, the failed knife is removed, a fresh throw is given). Limit Continue to 3 per run (config). Game Over also shows score, stage, best, and the buttons Continue (5 s countdown ring), Retry, Home and Store.

Difficulty rises fast. Stage 1 is a gentle tutorial, then every stage adds pressure: knives to throw = 6 + floor(stage/3) (max 16); pre-stuck knives from stage 2, growing quickly to 8; rotation speed starts ≈ 120°/s and rises ≈ 4°/s per stage (cap ≈ 330°/s). Stages are

generated from a seed and must always be solvable: validate that the free angular space can fit the knives to throw. All numbers live in difficulty.ts.

Obstacles and modifiers (introduced progressively; a stage may combine two or three):

1. Spikes embedded in the target (from stage 4): touching one fails the throw.
2. Golden apples (from stage 3, +10 apples) and crystal apples (from stage 8, +1 gem).
3. Speed patterns (from stage 2): constant, reverse every N seconds, sine pulse, stop-and-go, accelerate.
4. Changing targets (from stage 10): the target pulses in size, swaps its spin direction when a knife sticks, or shrinks as knives stick.
5. Mixed knives (from stage 6): the queue holds small, normal and wide knives that need different gaps; show the next knife type in the queue.
6. Orbiting shield arcs (from stage 12): barrier arcs circle the target and block the throw path when aligned, so the player must time the throw.
7. Drifting spikes (from stage 15): spikes that slowly slide around the target.
8. Ghost obstacles (from stage 20): spikes that fade in and out.

Bosses: every 5th stage in every mode is a boss fight. Each boss has a unique look, a name banner (BOSS FIGHT), a harder pattern, more knives and a short intro. Design at least 20 distinct bosses (food, objects and fantasy creations, no people or real animals), for example Giant Pumpkin, Ice Crystal, Iron Shield, Pizza Wheel, Gear Golem, Lava Core, Voodoo Mask, Frost Moon, Cactus King, Royal Crown, Dragon Egg, Cyber Core, Burger Beast, Sushi Roll, Candy Swirl, Treasure Chest, Disco Ball, Alien Orb, Ancient Idol, Clockwork Heart. How many bosses a run meets depends on the stage reached and the mode: Endless cycles through them with growing difficulty; Fixed Run uses the first six. Rewards: +25 apples and +3 gems on every win, and on the first win over each of the first 10 bosses, that boss's exclusive knife.

Game modes (Mode select screen; each card shows the player's best result):

1. **Endless:** infinite stages; the score is the stage reached; best saved.
2. **Fixed Run:** a set number of stages (default 30, config) ending with a final boss and a victory screen with rewards; replayable.
3. **Daily Challenge:** one seeded set of 10 stages per calendar day (the seed comes from the device date, so no server); one free attempt a day plus an extra attempt through a rewarded ad; rewards in apples and gems scale with stages cleared; a day-streak counter.
4. **Time Attack:** a 60 s clock; clearing a stage adds 8 s, a boss adds 15 s, a collision costs 5 s and resets that stage's knives; the score is stages cleared; a rewarded ad adds +15 s once per run; best saved.

Power-ups. Playing earns points (+10 per knife stuck, +25 per apple, +50 per golden apple, +15 for a close call: a knife landing within 2° of another without touching). Points fill a power meter; each time it fills, the player earns one power-up charge (the threshold grows with the stage). The player can also watch a rewarded ad to get a power-up of their choice (once per stage). Power-ups: Slow Motion (rotation ×0.4 for 5 s), Freeze (the target stops for 3 s), Aim Guide (highlights the safest gap for 3 throws), Extra Knives (+2 for this stage), Shield (absorbs one collision), Clean Sweep (removes up to 3 stuck knives). Availability grows with the stage: the first unlocks at stage 3, another about every 5 stages, and the number of charges allowed per stage grows too (1 early, 2 on boss stages). Power-up buttons sit on the screen edges, outside the tap-to-throw area.

Achievements and Daily Missions

- About 40 achievements with Bronze, Silver and Gold tiers (apples sliced, stages reached per mode, bosses defeated, knives owned, Time Attack scores, daily streaks, close calls). Each shows progress and a Claim button that pays apples or gems.
- 3 Daily Missions that refresh at local midnight (for example slice 20 apples, clear 8 stages, defeat a boss, use 2 power-ups, play the Daily Challenge). Each pays apples or gems; finishing all 3 opens a bonus chest. A rewarded ad can double a claimed apple reward.

Game feel (important): knife stick wobble (damped oscillation), wood-chip and juice particles, 40 ms hit-stop on stick, screen shake on fail and boss win, floating +N texts, Perfect and Close call pop-ups, haptics through @capacitor/haptics (light on stick, medium on apple, heavy on fail, success on boss win), and automatic pause when the app goes to the background.

Part 2 — Store and economy

Two currencies, medium economy. Apples are the common currency; Gems are rare and cannot be bought with real money (no in-app purchases in this version). Put every number in economy.ts so it is easy to tune. The targets below aim for a medium pace: a Common knife after one or two sessions, an Epic after a few days of play, a Legendary after a couple of weeks.

- **Apples:** +2 per apple sliced, +10 per golden apple, +1 per stage cleared, +25 per boss win, plus the Daily Gift, missions, achievements, the Daily Challenge and rewarded ads (+50 per ad, with a 2-minute cooldown on the Free apples button).
- **Gems:** +3 per boss win, +1 per crystal apple, Gold achievements, the Daily Mission chest, Daily Challenge streak milestones and Day 7 of the Daily Gift. Ads never double gems.
- **Rewarded ads always pay apples.** They never unlock a knife directly: if the player cannot afford a knife, its card offers Watch ad (+50 apples).

Daily Gift: a 7-day cycle. Days 1–5 give 10, 20, 30, 40 and 50 apples, Day 6 gives 75 apples, Day 7 gives 150 apples and 5 gems. The streak resets if a day is missed. Add a Double it (rewarded ad) button for the apples part and a claim animation. There is no half-hourly gift.

Store screen (spend the most polish here)

- Header with animated apples and gems balances and a Free apples button (rewarded ad). Sticky tabs: All, Apples, Gems, Boss, Owned.
- 3-column card grid. Each card has a rotating knife preview, name, rarity frame color plus icon (Common, Rare, Epic, Legendary, with a shimmer on Epic and Legendary), and a state chip: Equipped, Owned, the price in apples and/or gems with icons, or Defeat Boss N (locked, greyed). NEW badge for fresh unlocks.
- Tapping a card opens a preview modal: large rotating knife, collection, rarity, short flavor text and the action (Buy with apples, Buy with gems, Equip, or the lock requirement). If the player lacks currency, offer a rewarded ad for apples and show how many gems are missing. Add purchase animations (burst, toast). The equipped knife shows immediately in game and on Home.
- Knives are cosmetic only: they never change gameplay.
- Collection progress bars (for example Kitchen 3/5) with a one-time bonus of apples or gems for completing a set.

Knife catalog: exactly 50 knives in data/knives.ts, drawn procedurally with varied blade shapes, guards, handles and palettes (Legendary knives get a glow or particle trail). Common and Rare cost apples; Epic can be bought with either apples or gems (the player chooses); Legendary cost gems only; boss exclusives are won by defeating that boss for the first time.

Collection	Count	Rarity	Apples	Gems
Starter (Classic Steel)	1	Common	Free	—
Kitchen	5	Common	100–250	—
Pirate	4	Rare	300–450	—
Hunter	5	Rare	350–600	—
Ninja	5	Epic	700–1000	20–30
Ice	4	Epic	800–1100	25–35
Cyber Neon	4	Epic	900–1200	25–35
Voodoo	4	Epic	1000–1300	30–40

Collection	Count	Rarity	Apples	Gems
Cosmic	4	Legendary	—	50–70
Royal	4	Legendary	—	70–90
Boss exclusives	10	Epic / Legendary	Not for sale	—

Settings: separate Sound and Music toggles, haptics, reduce motion, Privacy settings (reopens the consent form where required), reset progress (double confirm), version.

Part 3 — Ads (real AdMob, Android only) and Capacitor

Use a maintained Capacitor AdMob plugin (for example @capacitor-community/admob or @capgo/capacitor-admob) that supports all four formats: banner, interstitial, rewarded and App Open. Check its current docs for the exact API; if no plugin supports a format, use the closest equivalent and tell me. Wrap everything in an AdService interface with two implementations: AdMobService (native) and MockAdService (web preview: a fake banner bar, and fake full-screen App Open, interstitial and rewarded overlays with a 3–5 s timer and a close button), selected with Capacitor.isNativePlatform(). The whole game must be testable in the Lovable preview.

The four ad types and their rules (config/adPolicy.ts):

1. **App Open:** every time the player opens the game: on each cold start after the splash, and when returning from the background after at least 15 s away. Never while another full-screen ad is showing, with a 30 s global cooldown; if the player was mid-run, show the Pause menu first.
2. **Banner:** adaptive. You choose the best position per screen (top or bottom) so it never covers a primary button, the HUD, the knife or the tap-to-throw area, and reserve a fixed slot so the layout does not jump. Store the choice per screen in adPolicy.ts and explain each choice in a short comment. During active play either hide it or keep it in a reserved strip outside the play area.
3. **Rewarded:** only when the player taps an opt-in button; the reward is granted only in the earned-reward callback. If no ad is ready, show a brief No ad available state and never freeze. Placements: Continue after a fail, Double stage and boss rewards (apples only), Free apples (Home and Store), Watch ad for apples on any knife the player cannot afford, Double Daily Gift (apples only), Double a claimed mission or achievement reward, a power-up of choice (once per stage), an extra Daily Challenge attempt, and +15 s in Time Attack.
4. **Interstitial between stages and games:** after every 3rd cleared stage (Endless and Fixed Run), and after every loss or run end in every mode, shown when the player leaves Game Over. Minimum 30 s between any two full-screen ads (config), never right

after a rewarded ad, never during play.

Preload all ads, reload after dismissal with exponential backoff, and fail silently without blocking gameplay. Mute audio during full-screen ads and restore it afterward.

Ad IDs: keep them only in src/config/ads.ts as clearly marked placeholders I will fill (Android only: the AdMob app ID plus banner, interstitial, rewarded and App Open unit IDs). Use test ads in dev builds through an env flag.

Compliance (all ages = mixed audience): show a neutral age screen once on first launch (for example a birth-year picker, worded neutrally, with nothing that hints older players get a better experience) and save the result on the device. Under 13, or if unknown: send child-directed requests (tagForChildDirectedTreatment = true, tagForUnderAgeOfConsent = true, maxAdContentRating = G, non-personalized ads only). Ages 13–17: maxAdContentRating = T. 18 and over: no content cap. Run the Google UMP consent flow for users aged 13 and over before their first ad request, and keep a Privacy settings entry to reopen it. Never put a tappable ad where a child could hit it by accident, and never use buttons that look like game controls to trigger ads.

Capacitor (Android only): add capacitor.config.ts (appId and appName from the chosen game name, webDir dist) and the plugins @capacitor/app, haptics, preferences, status-bar, splash-screen and screen-orientation. Lock portrait with screen-orientation, hide the status bar, handle the Android back button (Pause, then a Home confirm), and use env() safe-area insets.

Part 4 — Visual and audio direction

Aim for a premium, expensive-looking result at the level of top-grossing casual games: depth, layered lighting, rich gradients, soft shadows, rim highlights and polished micro-animations on every touch. Nothing may look like a flat prototype.

- **Worlds:** the background changes with the stages. Every 5 stages (ending with a boss) form a world with its own palette, target material and layered parallax background (gradient sky, mid-layer silhouettes, foreground ambient particles). 12 worlds: Wood Workshop, Orchard, Frozen Cavern, Desert Canyon, Volcano, Neon City, Haunted Manor, Pirate Cove, Deep Space, Candy Land, Ancient Temple, Royal Palace. Endless cycles through them with shifted palettes, with a smooth cross-fade between worlds.
- **UI:** glossy, chunky buttons with inner highlights and a press animation; pill-shaped currency counters with animated icons; modals that bounce in; count-up numbers; confetti and glow on rewards; shimmer on Epic and Legendary frames; one consistent icon set drawn in code. Type: a chunky display font for headings with outline and shadow, and a clean rounded sans for body text, both bundled locally.

- **Knives and targets:** every knife has a blade gradient with a specular streak, an edge highlight, a detailed guard and handle, and a soft drop shadow; Legendary knives glow and leave a trail. Targets show material detail (wood rings, ice, stone, candy) and break into physical pieces. Bosses have idle animations and expressive hit reactions.
- **Big moments:** camera punch on a boss entrance, slow motion on the final knife of a boss, a Boss Defeated celebration with the new knife reveal, and screen transitions of 200–300 ms.
- **Audio:** one music loop per world (a generative WebAudio loop is fine, built so real audio files can replace it later) and a full sound-effect set (thwack, apple slice, fail, shatter, boss intro, UI taps, coin count, power-up). Separate Music and Sound toggles in Settings and on Home, both saved. Mute audio during ads.
- **Accessibility:** a reduce-motion option, and rarity shown by icon as well as color.

Quality bar

- No lorem ipsum and no dead buttons; every screen has transitions and pressed states.
- Stable 60 fps on mid-range phones, cleanup on screen unmount, strict TypeScript, and unit tests for the collision/angle logic and stage solvability.
- Keep the code modular so economy, difficulty and ad policy can be changed only in config files.

After Lovable finishes: your Android checklist (Google Play only)

1. Sync the project to GitHub and clone it to your computer.
2. Run npm install, npm run build, npx cap add android, then npx cap sync, and open the android folder in Android Studio.
3. Put your real AdMob app ID in AndroidManifest.xml (meta-data com.google.android.gms.ads.APPLICATION_ID) and the four ad unit IDs in src/config/ads.ts.
4. Confirm the package name Lovable chose before the first upload: it cannot be changed after publishing on Google Play.
5. Test with test ads first. Never tap your own live ads: it can get your AdMob account suspended.
6. In Play Console, set Target audience and content to the age groups you want (all ages includes children, so Google's Families Policy applies), declare that the app contains ads, fill in Data safety and the content rating questionnaire, and add a privacy policy URL. I am not a lawyer: read Google's current Families Policy before submitting.
7. In AdMob, link the app to its Play listing and set up the consent message under Privacy & messaging.

8. Generate icons and splash screens with @capacitor/assets, build a signed AAB in Android Studio, and release to the internal testing track first.

Defaults I chose (tell me to change any)

- Fixed Run: 30 stages (6 bosses). Time Attack: 60 s start, +8 s per stage, +15 s per boss, −5 s per collision.
- Continue (rewarded ad): max 3 per run and it resumes the same stage; any other failed run restarts at stage 1.
- Free apples ad: +50 apples with a 2-minute cooldown.
- Minimum gap between full-screen ads: 30 s; App Open after at least 15 s away.
- Your answer about unlocking knives with points is read as Gems. Rewarded ads give apples and never unlock a knife directly.
- 12 worlds of 5 stages; 20 bosses; the first 10 bosses give an exclusive knife.
- Age screen: under 13 child-directed, 13–17 rating T, 18 and over uncapped.